pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Spread Duration and DTS

WEEK 3 · DAY 3

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The desk lead's note: a copy error?

The Desk Lead's Note

Two more columns of our month-end file: "spread_duration" and "dts". In every one of the 200 rows "spread_duration" is the same number as "duration". A copy error? Rebuild both columns from the bond math, and tell me how the book's DTS splits by rating bucket.

The issuers are invented; every number describes the data.

Read the Book

```
spread_duration equals duration in 200 of 200 rows
```

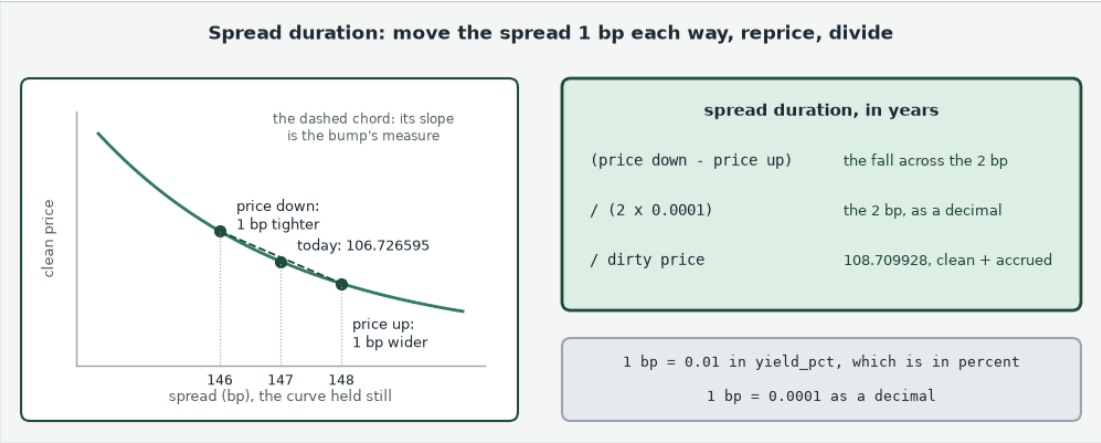
cusip	issuer	oas	duration	spread_duration	dts
99080CAE8	Bezane Finance	147	4.627	4.627	680.2
99049XAE2	Bezamyth Machinery	535	2.165	2.165	1158.3
99039NAC0	Bezoquil Freight	153	3.250	3.250	497.2
99039NAG1	Bezoquil Freight	218	12.113	12.113	2640.6
99034IAD4	Bezorin Finance	127	7.338	7.338	931.9

```
.................................   [100%]
35 passed in 0.50s
```

"oas" is the spread notebook 12 rebuilt from the curve, to 2.3e-13 bp.

Measure one bond's spread duration first, then answer the question about the copy.

13.1 Bump the spread, reprice




```
dirty = bondmath.dirty_price(settlement, maturity, coupon_pct, yield_pct)
spread_duration_years = (price_down - price_up) / (2 * 0.0001) / dirty
dirty price 108.709928; spread duration 4.626971 years
```

Spread duration is the percent the price moves for 100 bp of spread, in years. It is measured the way notebook 09 checked DV01: move the input 1 bp each way and reprice. The file's yield is the curve plus the spread, so with the curve held still, a move in the spread moves the yield.

```
Bezane Finance: 7.0 percent coupon, maturing 2032-06-18
yield 5.606 percent, spread 147 bp
```

Accrued interest does not move with the spread, so the clean prices' difference is the dirty prices' difference.

Excel side by side. Tested in Excel for Bezane, one cell each.

Job	Excel
Price at the yield (B1)	=PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100,100,2,0)
Spread 1 bp tighter (B2)	=PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100-0.0001,100,2,0)
Spread 1 bp wider (B3)	=PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100+0.0001,100,2,0)

13.2 Hold the curve: a spread move is a yield move

```
Accrued (B4)
=100*7/100/2*COUPDAYBS(DATE(2026,9,30),DATE(2032,6,18),2,0)/COUPDAYS(DATE(2026,9,30),DATE(2032,6,18),2,0)
Spread duration (B5)
=(B2-B3)/(2*0.0001)/(B1+B4)
```

Inside "PRICE" the yield is a decimal, so 1 bp is "0.0001" there and in the slope alike.

Bezane's price moves about 4.63 percent of its value for 100 bp of spread: spread duration is 4.626971 years.

Notebook 12 found the file's rule: yield = curve yield + spread / 100. Hold the curve, and 1 bp more spread is 1 bp more yield.

The same price to twelve decimals. The same with "147-1" in B10 gives "=B10-B2" of 0.

```
import math
years = bondmath.years_to_maturity(settlement, maturity)
curve_pct = round(3.55 + 1.05 * (1 - math.exp(-years / 7)), 3)
price_wider = bondmath.price(settlement, maturity, coupon_pct, curve_pct + (spread_bp + 1) / 100)
4.136 + 147 / 100 = 5.606 percent
```

Job	Excel
Curve plus spread, 1 bp wider (B11)	=PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,(4.136+(147+1)/100)/100,100,2,0)
Against the yield moved	=B11-B3

13.3 spread_duration, docstring first

```
%%writefile -a bondmath.py
def spread_duration(settlement: date, maturity: date, coupon_pct: float, yield_pct: float, frequency: int = 2,
                    basis: str = "30/360", bump_bp: float = 1.0) -> float:
    if bump_bp <= 0:
        raise ValueError(f"bump_bp must be positive, not {bump_bp}")
    bump_pct = bump_bp / 100
    bump_decimal = bump_bp / 10000
    price_spread_down = price(settlement, maturity, coupon_pct, yield_pct - bump_pct, frequency, basis)
    price_spread_up = price(settlement, maturity, coupon_pct, yield_pct + bump_pct, frequency, basis)
    full_price = dirty_price(settlement, maturity, coupon_pct, yield_pct, frequency, basis)
    return (price_spread_down - price_spread_up) / (2 * bump_decimal) / full_price
```

"bump_pct" moves "yield_pct", which is in percent; "bump_decimal" divides the slope. Two names for one bump keep the two units apart. A bump of zero would divide by zero, so the function refuses it.

```
bondmath 4.626970799151, Excel 4.626970799150
gap 1.3e-12 years
```

```
200 bonds; Bezane 4.626970799150 years
```

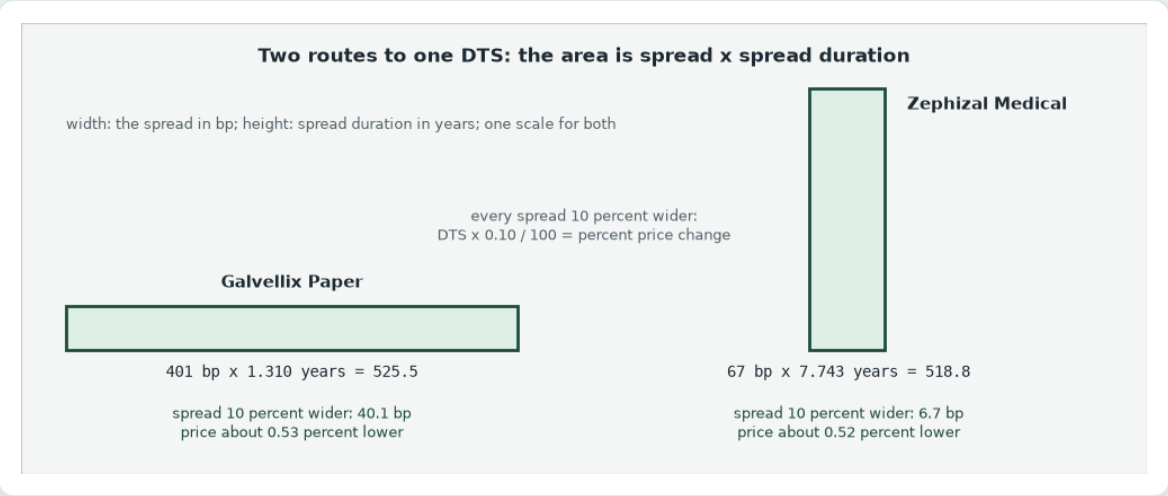
Excel side by side: the docstring's formula in one cell

```
=(PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100-0.0001,100,2,0)-PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100+0.0001,100,2,
0))/(2*0.0001)/(PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100,100,2,0)+100*7/100/2*COUPDAYBS(DATE(2026,9,30),DATE(2032,6,18),2,
0)/COUPDAYS(DATE(2026,9,30),DATE(2032,6,18),2,0))
```

```
.................................   [100%]
36 passed in 0.50s
```

13.4 DTS: spread duration times spread

Spread duration says what a move of so many bp does to the price. Spreads do not all move by the same number of bp. DTS, duration times spread, measures a move stated as a share of the spread: spread duration in years times the spread in bp. It is a published measure: Ben Dor, Dynkin, Hyman, Houweling, van Leeuwen and Penninga, "DTS (Duration Times Spread)", *Journal of Portfolio Management*, Winter 2007 (CFA Institute's summary). On credit index data from 1989 to 2005 they found that bonds at wider spreads moved more in bp, roughly in proportion to their spread. If a spread widens by 10 percent of itself, the move is 0.10 x spread bp, so the price changes by about -DTS x 0.10 / 100 percent.

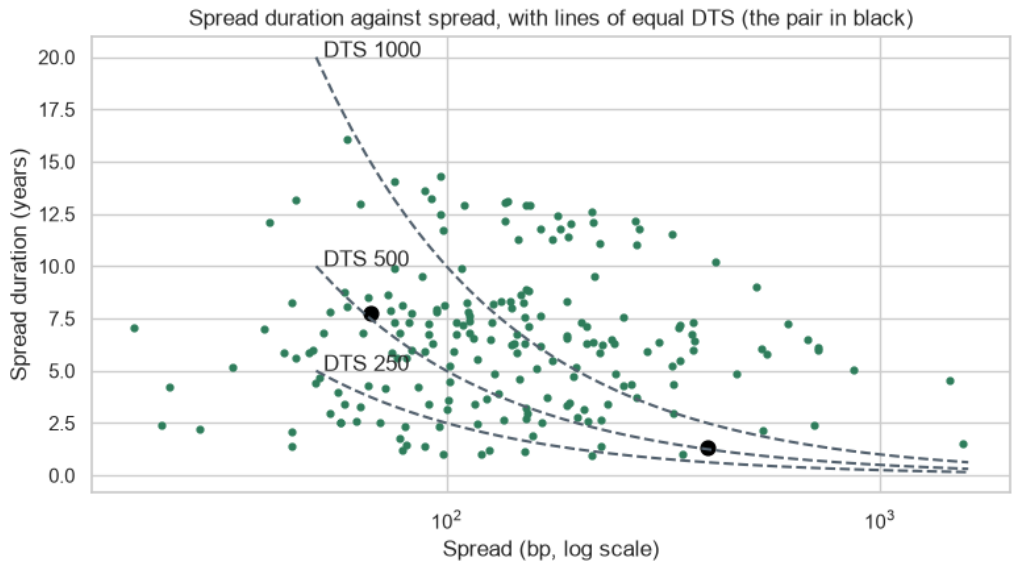


Two bonds of the book reach almost the same DTS by opposite routes. With every spread 10 percent wider, both lose about half a percent.

cusip	issuer	oas	spread_duration_bm
99088KAA0	Galvellix Paper	401	1.310
99062KAE2	Zephizal Medical	67	7.743

Lines of equal DTS, and the dts function

A line of equal DTS is spread duration = DTS / spread. "ax.set_xscale('log')", as in notebook 01, spaces the spread axis by multiples: 50 to 100 bp is as wide as 500 to 1000.



A bond on a dashed line has that DTS, however it got there: short and wide, or long and tight.

```
%%writefile -a bondmath.py
def dts(spread_duration_years: float, spread_bp: float) -> float:
    return spread_duration_years * spread_bp
```

One line, on purpose: the docstring carries the unit, years times bp, so the number is never read as a percent or a dollar amount.

```
bondmath.dts 680.1647; the file's dts 680.2
```

The file rounds its "dts" to one decimal.

Job	Python	Excel	Result
DTS (B7)	bondmath.dts(bezane_sd, spread_bp)	=B5*147	680.1647074750472

Two bonds with the same DTS lose about the same share of value when every spread moves by the same share, whatever their duration and spread.

13.5 The file's two columns, rebuilt

The file's generator, "tools/make_holdings.py", wrote "spread_duration" as its "duration", modified duration rounded to 3 decimals, for 13.2's reason. Its "dts" is that rounded number times "oas", rounded to 1 decimal.

```
gap_sd = (holdings['spread_duration_bm'] - holdings['spread_duration']).abs()
assert (gap_sd <= 0.0005 + 1e-9).all(), "a bond's spread duration differs from the file"
largest gap 0.000498707 years
```

```
from_file = (bondmath.dts(holdings['spread_duration'], holdings['oas']) - holdings['dts']).abs()
largest gap 0.0500000000002
```

```
holdings['dts_bm'] = bondmath.dts(holdings['spread_duration_bm'], holdings['oas'])
from_math = (holdings['dts_bm'] - holdings['dts']).abs()
largest gap 0.3999; 82 of 200 bonds above 0.05
```

A spread of 871 bp multiplies the file's 3-decimal rounding of spread duration, so no tolerance is claimed for this route.

Five rows typed in, with the same five CUSIPs as notebook 12's test.

```
%%writefile -a test_bondmath.py
def test_dts_reproduces_file_rows():
    rows = [
        ("99080CAE8", 4.627, 147, 680.2),
        ("99049XAE2", 2.165, 535, 1158.3),
        ("99039NAC0", 3.25, 153, 497.2),
        ("99039NAG1", 12.113, 218, 2640.6),
        ("99034IAD4", 7.338, 127, 931.9),
    ]
    for cusip, spread_duration_years, spread_bp, file_dts in rows:
        assert bondmath.dts(spread_duration_years, spread_bp) == pytest.approx(file_dts, abs=0.05 + 1e-9), cusip
```

```
....................................    [100%]
37 passed in 0.50s
```

Both columns rebuilt: what differs is the file's rounding, never the math.

13.6 When did 10000 arrive? The pickaxe

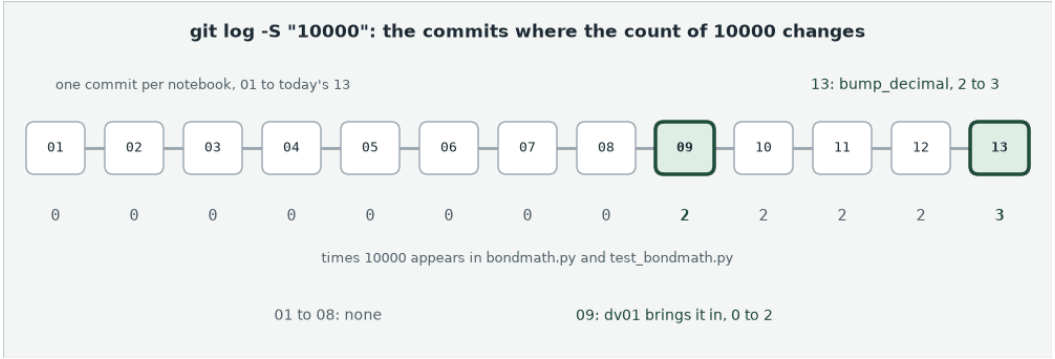
```
!git -C "{work}" diff --stat
subject = 'Notebook 13: spread_duration and dts'
body = 'Spread duration and DTS feed the risk report of notebook 14'
!git -C "{work}" add bondmath.py test_bondmath.py
!git -C "{work}" commit -q -m "{subject}" -m "{body}"
 bondmath.py    | 34 ++++++++++++++++++++++++++++++++++
 test_bondmath.py | 23 +++++++++++++++++++++++
 2 files changed, 57 insertions(+)
```

"bump_decimal" divides by 10000, and so did an earlier day. "git log -S "10000"" lists the commits that changed how many times the text "10000" appears in the files. Git calls it the pickaxe.

```
!git -C "{work}" log -S "10000" --oneline
e42ff20 Notebook 13: spread_duration and dts
5fcb7c6 Notebook 09: macaulay_duration, modified_duration, and dv01
```

```
!git -C "{work}" log -S "10000" --oneline -- test_bondmath.py
5fcb7c6 Notebook 09: macaulay_duration, modified_duration, and dv01
```

Only notebook 09: its DV01 test's comment holds the formula, and today's tests do not use the number.



Two commits, newest first: the two where the count changes. The oldest is where 10000 came in: notebook 09's "dv01", whose docstring checks it as "MDURATION(...)*(PRICE(...)+accrued)/10000". The search reads the changes, not the messages: neither subject mentions 10000.

"git log -S" finds the commit that brought a number in: 10000 arrived with "dv01" in notebook 09.

13.7 The book's DTS from your own functions

A bond's weight is its share of the book's market value; its contribution is the weight, as a fraction, times its DTS. The contributions add up to the book's DTS, as the Course 1 project's picture of three invented bonds showed. This time every DTS is your own.

```
total_value = holdings['market_value'].sum()
holdings['weight_pct'] = holdings['market_value'] / total_value * 100
holdings['dts_contribution'] = holdings['weight_pct'] / 100 * holdings['dts_bm']
book_dts = holdings['dts_contribution'].sum()
book DTS, the contributions added up: 1064.110539 years x bp
```

```
book DTS, the weighted average: 1064.110539 years x bp
book DTS from the file's dts column: 1064.113487 years x bp
```

Your own book DTS, two ways, to the sixth decimal. The file's rounded column moves it by 0.0029.

Excel side by side. The book in a sheet, rows 2 to 201: rating bucket in B, "oas" in F, market value in G

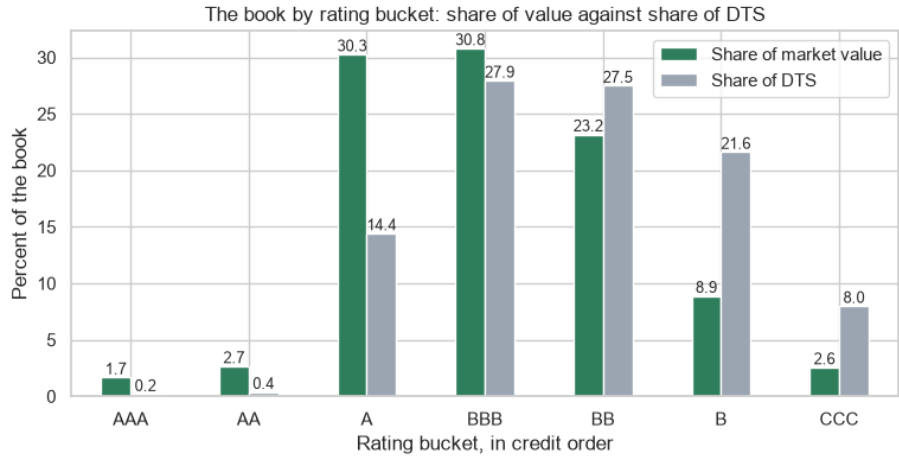
Job	Python	Excel	Result
Spread duration (L)	`bondmath.spread_duration(*bond)`	=(I2-J2)/(2*0.0001)/(H2+K2)	filled down
DTS (M)	`bondmath.dts(...)`	=L2*F2	filled down
Weight (N)	`market_value / total_value * 100`	=G2/SUM($G$2:$G$201)*100	filled down
Contribution (O)	`weight_pct / 100 * dts_bm`	=N2/100*M2	filled down
Book DTS	`holdings['dts_contribution'].sum()`	=SUM(O2:O201)	1064.1105392423183
Book DTS, one cell	`the weighted average`	=SUMPRODUCT(G2:G201,M2:M201)/SUM(G2:G201)	1064.1105392423187

The book by rating bucket, and by grade

bucket	weight_pct	dts_contribution	share_of_dts_pct
AAA	1.67	1.95	0.18
AA	2.66	4.00	0.38
A	30.26	153.45	14.42
BBB	30.80	297.24	27.93
BB	23.18	292.47	27.49
B	8.88	230.04	21.62
CCC	2.56	84.95	7.98

"=SUMIFS(O2:O201,B2:B201,"BBB")" returns 297.2402393040499, BBB's row; every bucket's "SUMIFS" equals the table to 1e-9.

grade	weight_pct	dts_contribution	share_of_dts_pct
High Yield	34.6	607.5	57.1
Investment Grade	65.4	456.6	42.9



From AAA to BBB each bucket's share of value is larger than its share of DTS; from BB down, the DTS share is larger.

A description of where the spread risk of this invented book sits, nothing more.

High Yield is 34.6 percent of the book's value and 57.1 percent of its DTS: the dollars and the spread risk sit in different buckets.

Bring it home, and what you can now do

Copy today's two files into your own "my_bondmath" folder, then test and commit there. The copy replaces your files: anything of your own must be in the work folder first.

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_13_k4w7q2xr"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py", "$work\.gitignore" .
python -m pytest -q
git diff --stat
git add bondmath.py test_bondmath.py
git commit -m "Notebook 13: spread_duration and dts" -m "Spread duration and DTS feed the risk report of notebook 14"
git log -S "10000" --oneline
git log -S "10000" --oneline -- test_bondmath.py
```

Your list may be longer: an exercise of your own that used 10000 is listed too. In Colab, download the two files from the work folder in the file browser. The work folder stays in your temp folder.

You Can Now

Measure spread duration by moving the spread 1 bp each way and repricing, with "spread_duration" or three "PRICE" cells.

Compute DTS with "dts" and say what it measures: the price change when a spread moves by a share of itself.

Rebuild the file's "spread_duration" and "dts" columns, and name the rounding that separates them.

Add up the book's DTS by bond and by rating bucket, with pandas or with "SUMPRODUCT" and "SUMIFS".

Find the commit that brought a number in, with "git log -S".

Predict first: 1 bp wider, and the bump

Predict 1

Bezane's spread widens 1 bp and the curve stays put. How far does its clean price fall, in points per 100 of face? Write down a number.

```
price_down = bondmath.price(settlement, maturity, coupon_pct, yield_pct - 0.01)
price_up = bondmath.price(settlement, maturity, coupon_pct, yield_pct + 0.01)
price_today = bondmath.price(settlement, maturity, coupon_pct, yield_pct)
print(f'1 bp tighter {price_down:.6f}, today {price_today:.6f}, 1 bp wider {price_up:.6f}')
print(f'the fall for 1 bp wider: {price_today - price_up:.4f} per 100')
```

Predict 2

A spread move is a yield move, so spread duration should be notebook 09's modified duration. The bump measures it between two prices 2 bp apart. For Bezane, the two are:

A) equal in every digit: a spread move is exactly a yield move

B) equal to about six decimals: the price curve bends a little over 2 bp

C) different in the second decimal: the bump is in basis points

```
modified = bondmath.modified_duration(settlement, maturity, coupon_pct, yield_pct)
print(f'the bump:          {spread_duration_years:.9f} years')
print(f'modified duration: {modified:.9f} years')
print(f'difference:        {spread_duration_years - modified:.2e} years')
```

Predict first: a bigger bump, and a doubled spread

Predict 3

Measure Sylvellix with bumps of 1, 5, 25 and 100 bp. Five times the bump does what to its gap above modified duration?

A) about the same: the gap is rounding

B) about five times: the gap grows with the bump

C) about twenty-five times: the gap grows with the square of the bump

```
sylvellix = bonds[furthest]
modified_syl = bondmath.modified_duration(*sylvellix)
bump_table = pd.DataFrame({'bump_bp': [1, 5, 25, 100]})
bump_table['gap_years'] = [bondmath.spread_duration(*sylvellix, bump_bp=b) - modified_syl
                           for b in bump_table['bump_bp']]
bump_table.style.format({'gap_years': '{:.1e}'}).hide(axis='index')
```

Predict 4

Bezane's spread doubles, 147 bp wider, and DTS says the price falls 6.80 percent. A full reprice says:

A) a smaller fall, by a few tenths of a percent

B) the same fall, to the hundredth

C) a larger fall, by a few tenths of a percent

```
dts_bezane = bezane_sd * spread_bp
print(f'DTS {bezane_sd:.6f} years x {spread_bp} bp = {dts_bezane:.4f} years x bp')
for share in [0.10, 1.00]:
    widening_bp = share * spread_bp
    new_yield = yield_pct + widening_bp / 100
    dirty_after = bondmath.dirty_price(settlement, maturity, coupon_pct, new_yield)
    print(f'{widening_bp:.1f} bp wider: DTS says {-dts_bezane * share / 100:.4f} percent, '
          f'a full reprice {(dirty_after - dirty) / dirty * 100:.4f}')
```

The note, answered

The note asked	Answer
A copy error?	no: a spread move is a yield move, so spread duration is modified duration
Both columns rebuilt	spread_duration within 0.0005 and dts within 0.05 of the file's own columns
DTS by rating bucket	BBB 27.9, BB 27.5 and B 21.6 percent of it; High Yield 57.1 percent on 34.6 of the value

Be proud: every number behind the desk's spread risk now comes from your own tested functions.

Next is Portfolio Aggregation and Scenarios: the 200 bonds rolled into one book and repriced under what-ifs.